POWER PULSE

# Overhead Line Insulators Important Formulae

Notes by BALAJI SIR

## 1 Voltage distribution in a string

1. $k = \dfrac{C_{shunt}}{C_{self}}$ (pin-to-earth/self capacitance), unit next to the cross-arm $= V_1$
2. $$V_2 = V_1(1+k), \qquad V_3 = V_1(1+3k+k^2)$$
   *3-disc string*
3. $$V_4 = V_1(1+6k+5k^2+k^3)$$
   *4-disc string*
4. $$V = V_1 + V_2 + V_3 = V_1(3+4k+k^2)$$
   *(3 discs)*
5. $$\text{String efficiency} = \frac{V_{string}}{n \times V_{\text{disc nearest the conductor}}} \times 100$$
   *always < 100%*
6. $$n = \frac{V_{ph}}{V_{disc} \times \eta_{string}}$$
   *number of discs*

## 2 Improving the string efficiency

7. Longer cross-arm ($k \downarrow$), grading ring, capacitance grading
   *the disc nearest the line has the largest voltage*
8. Grading ring: adds capacitance from the metal fittings to the line

## 3 Insulator ratings and types

9. $$\text{Safety factor} = \frac{\text{puncture voltage}}{\text{flash-over voltage}}$$
   *puncture strength should exceed flash-over*
10. Wet flash-over $\approx 0.8 \times$ dry flash-over
11. Pin: up to 33 kV; suspension: above 33 kV; strain: dead ends; shackle: LV
12. Number of discs (approx.): 11 kV–1, 33 kV–3, 66 kV–5, 132 kV–9, 220 kV–14
    *typical values*